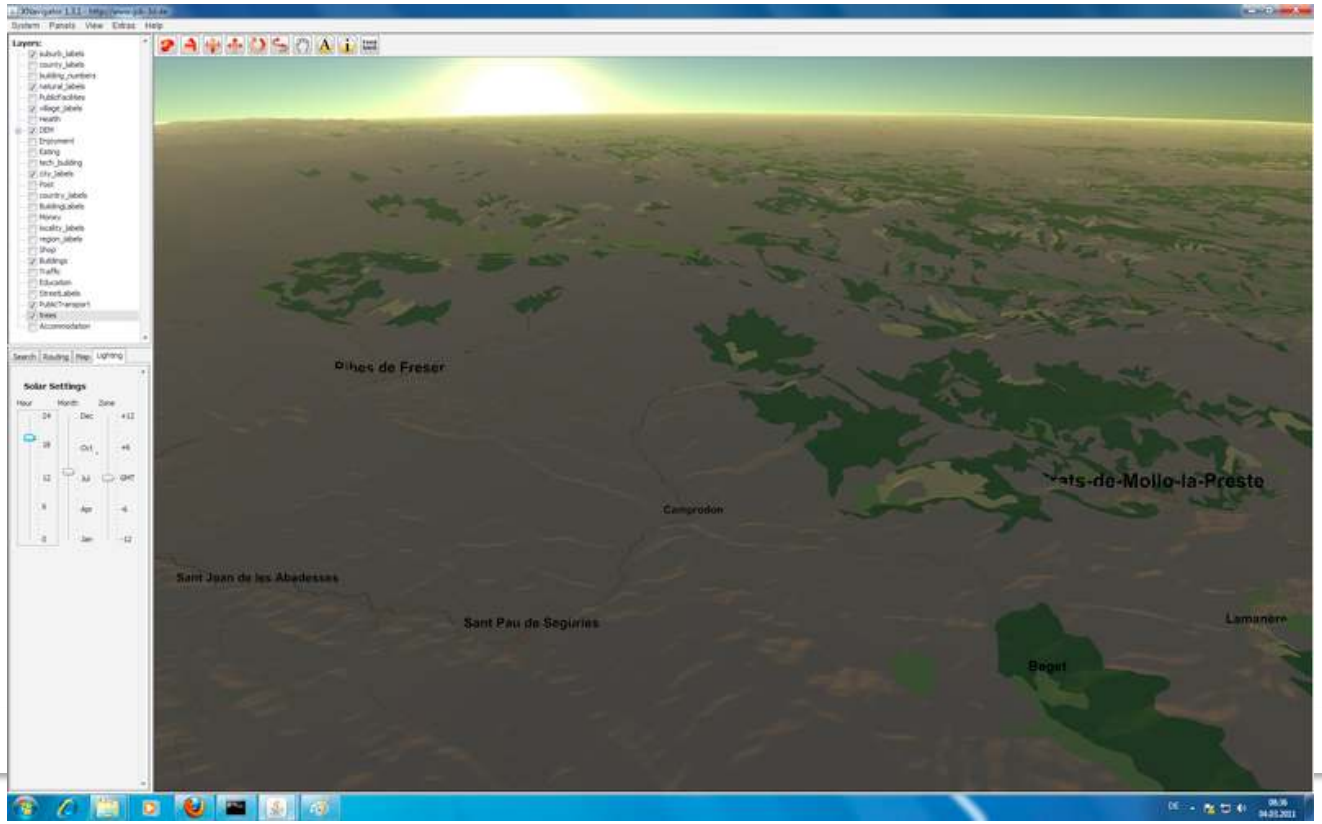
From OSM.planet to planet.OSM

www.osm-3d.dorg



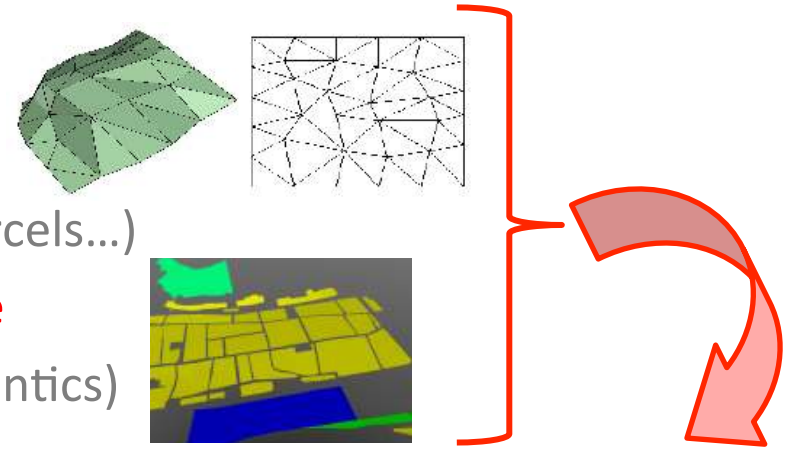
Alexander Zipf
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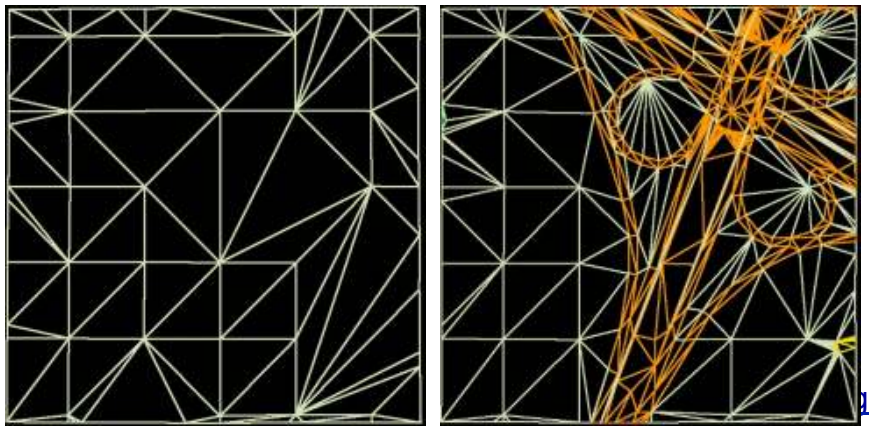


Generating 3D landscape models

Combine heterogeneous data:

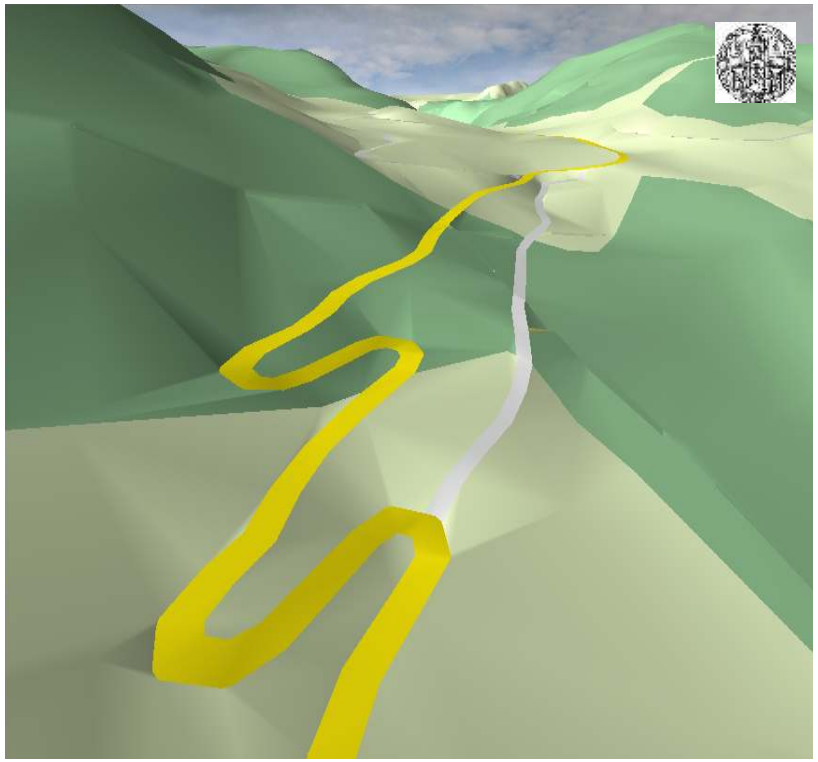
- 3D buildings (city model)
- digital elevation models (DEM)
  - triangulated 3D points
- Landuse (streets, water, forest, parcels…)
  - 2D vector data with landuse type
  - aerial / satellite images (no semantics)
- other objects
  - trees, signs, street furniture
  - landmarks, POIS, …

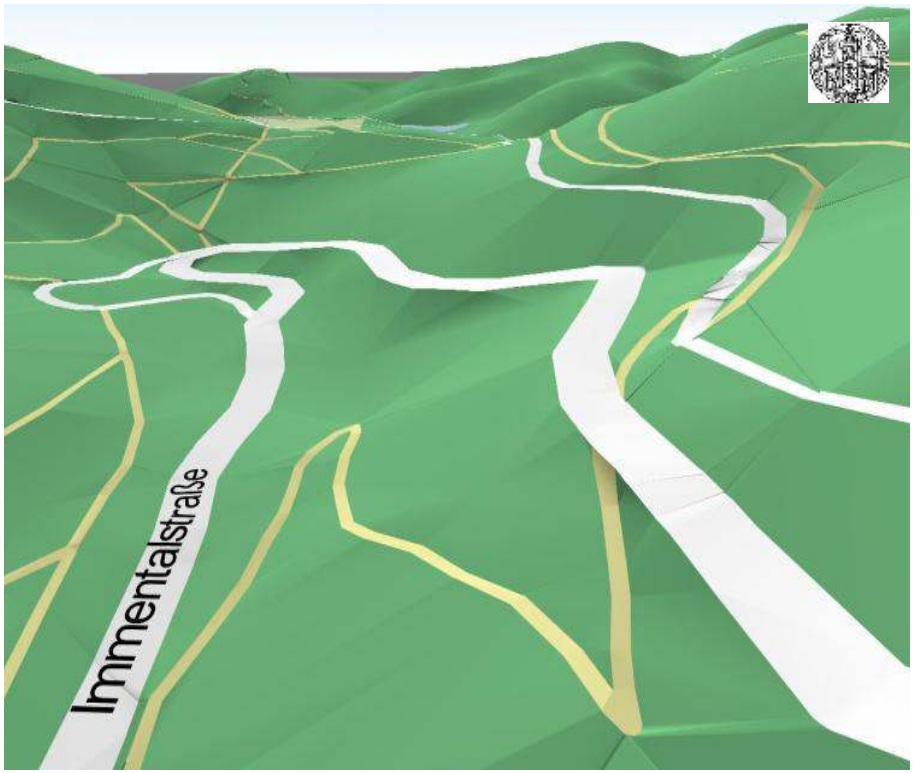




Smoothed streets in 90 meter DEM

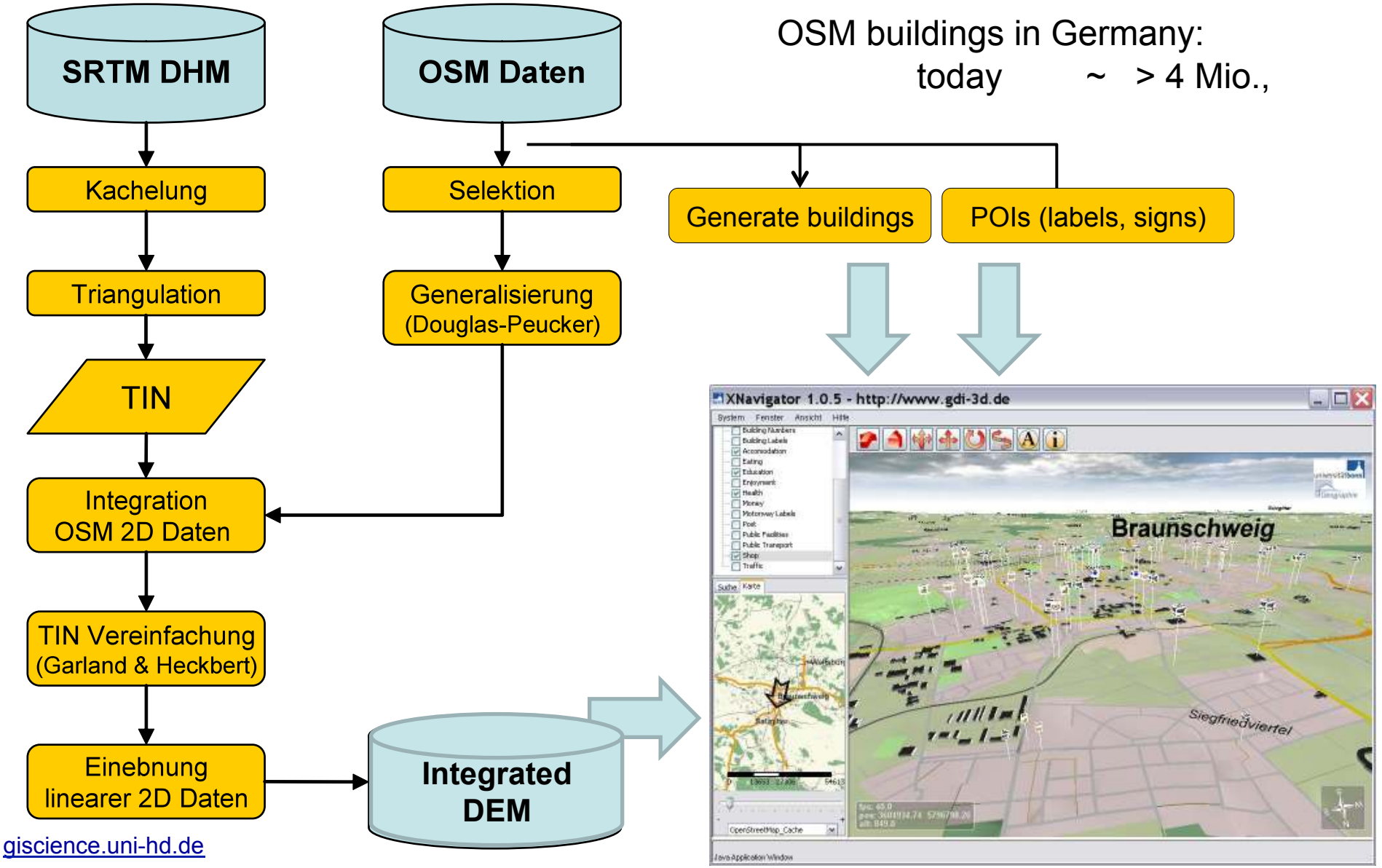
- Adaptive flattening of streets improves original coarse DEM





(energy function dependend from land use moves triangles)

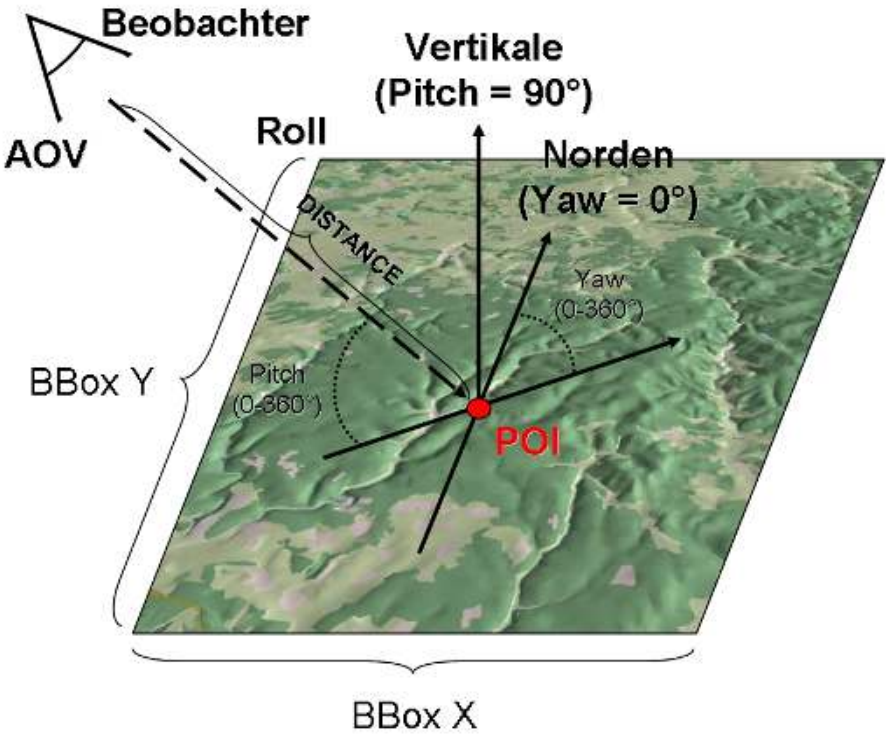
OSM-3D.org: OSM Landuse & SRTM DGM



giscience.uni-hd.de

OGC Web 3D Service (W3DS) draft

- delivers 3D scenes
  - display elements
  - VRML, X3D, KML, Collada…
- imports 3D city models
- defined http interface

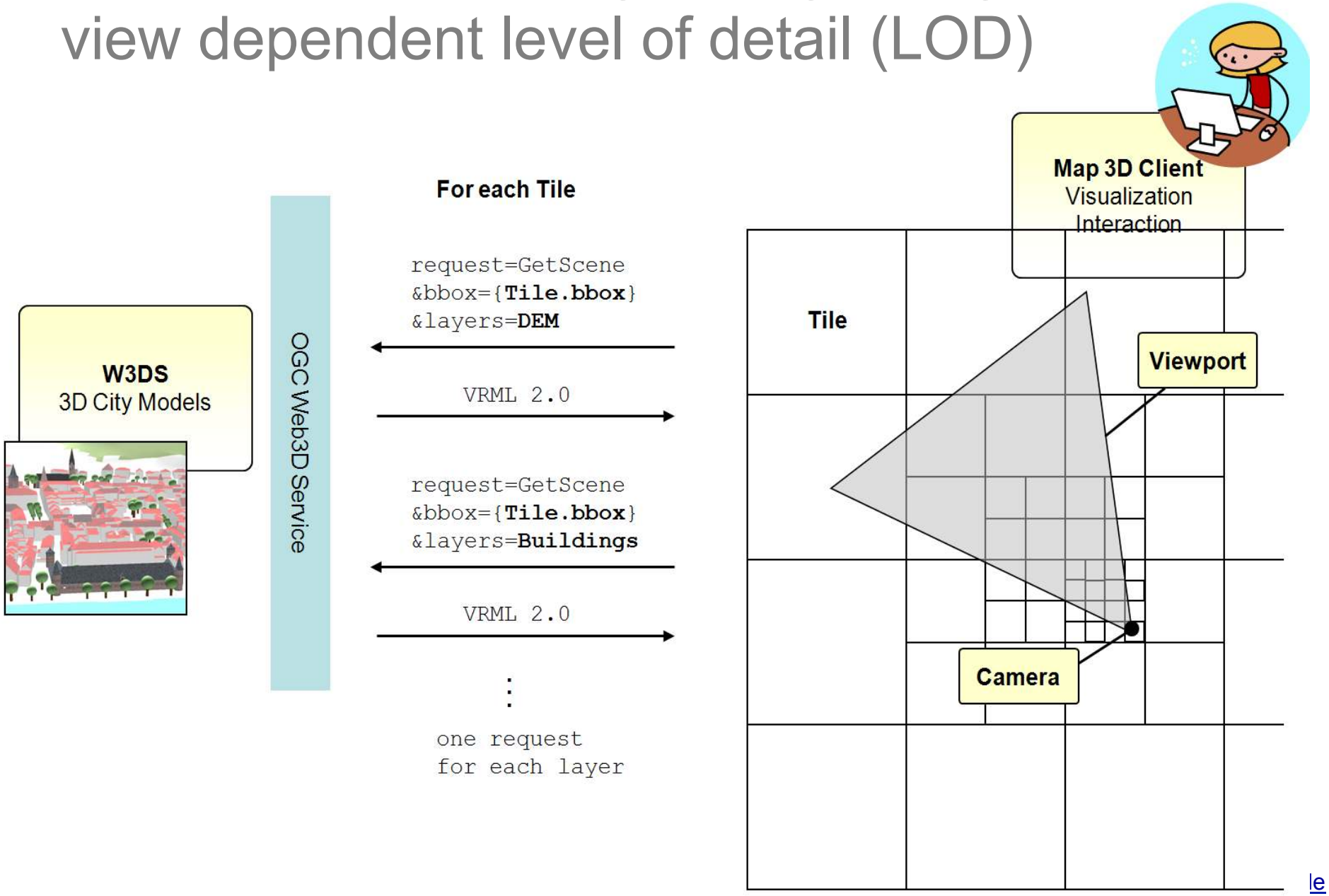


http://myserver/w3ds?
request=GetScene&
bbox=3448244,5540223,3448582,5540495&
poi=3448410,5540385,0&
layers=trees,litfass,laterne,ampel,buildings,streets,rhein&
styles=tree,litfass,laterne,ampel,bisque,black,lake

&aov=45&pitch=45&roll=20&yaw=270&distance=80& ..

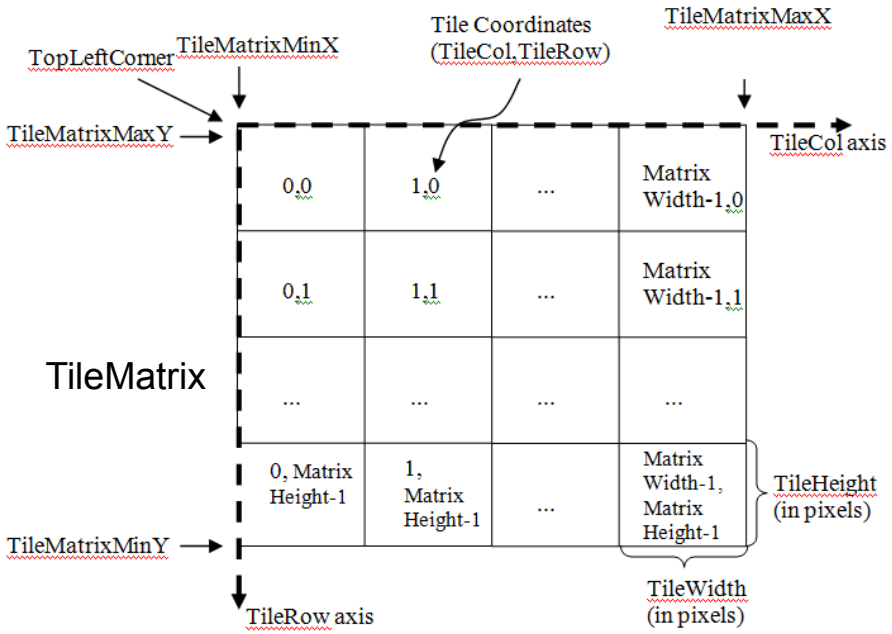
zipf@uni-heidelberg.de

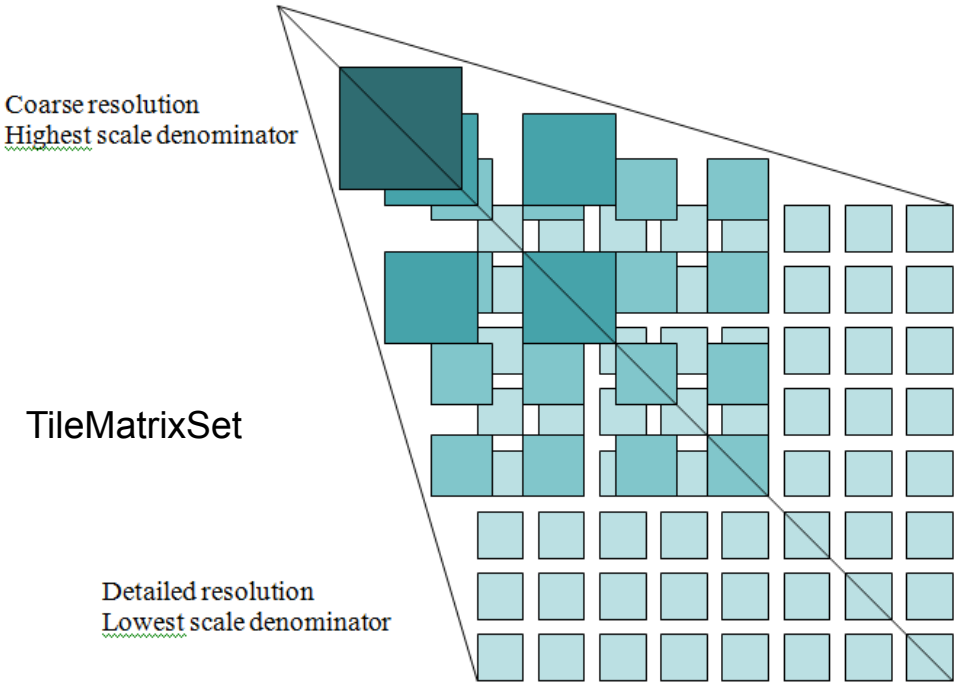
Support for streaming through tiling & view dependent level of detail (LOD)



New OGC W3DS Spec draft: GetTile-Operation

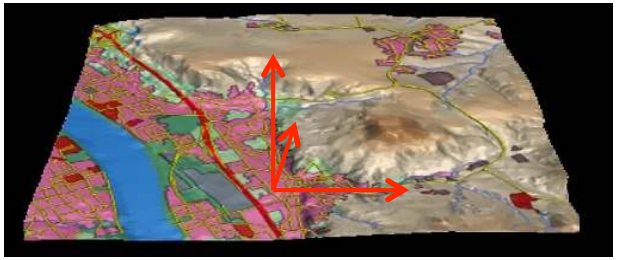
- Include definitions of TileMatrix and TileMatrixSet



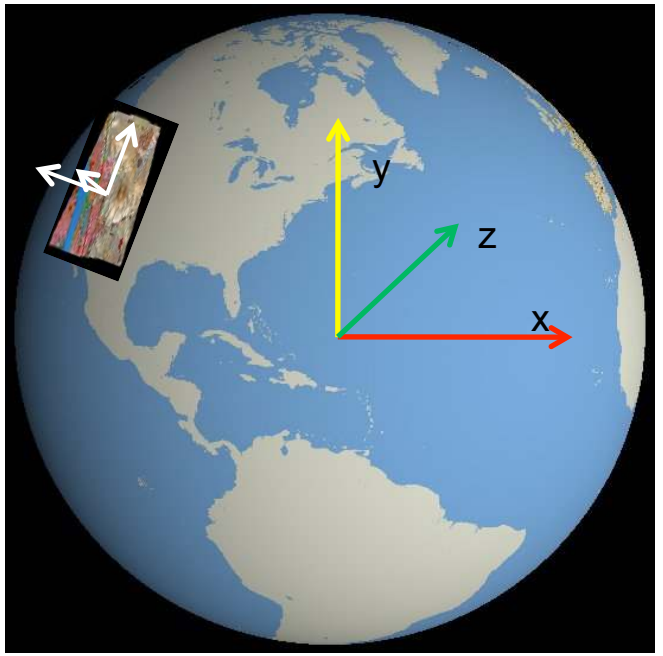


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Scenegraph Transformations



Data as received from W3DS
in Map Projection CRS
(Spherical Mercator)



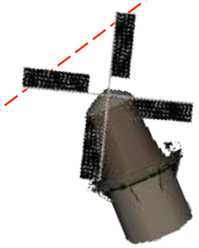
Data as displayed in the final
application in Cartesian
Coordinates

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Scenegraph Transformations



Shape in local Coordinates



Shape in local Coordinates

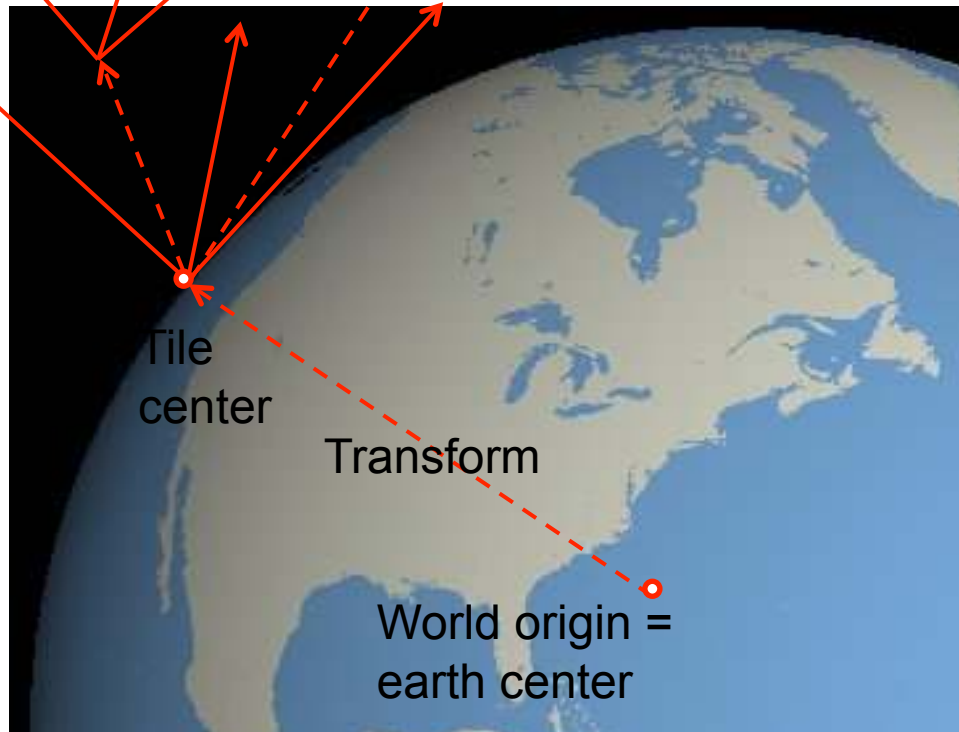
Tile center

Transform

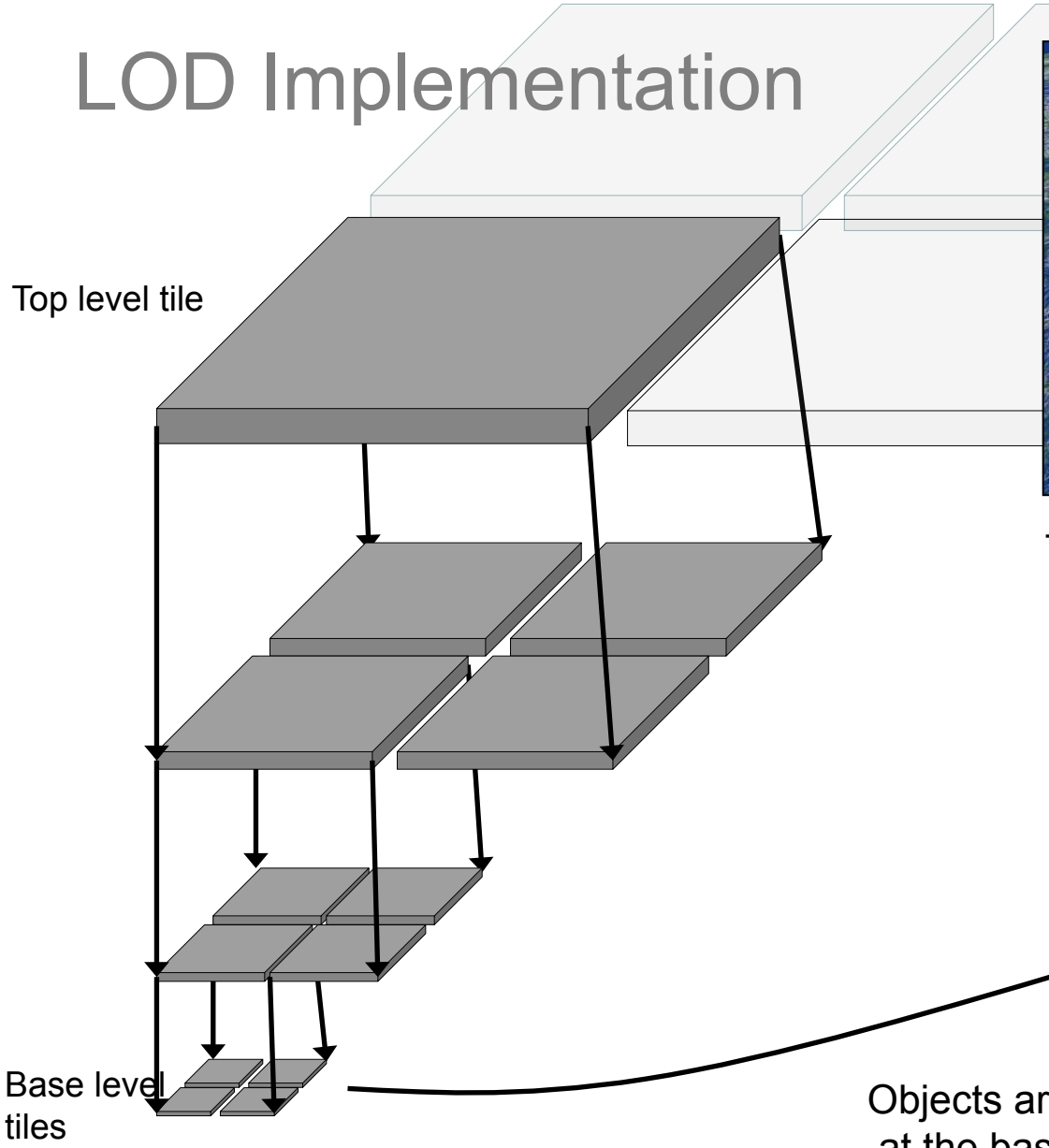
Map origin

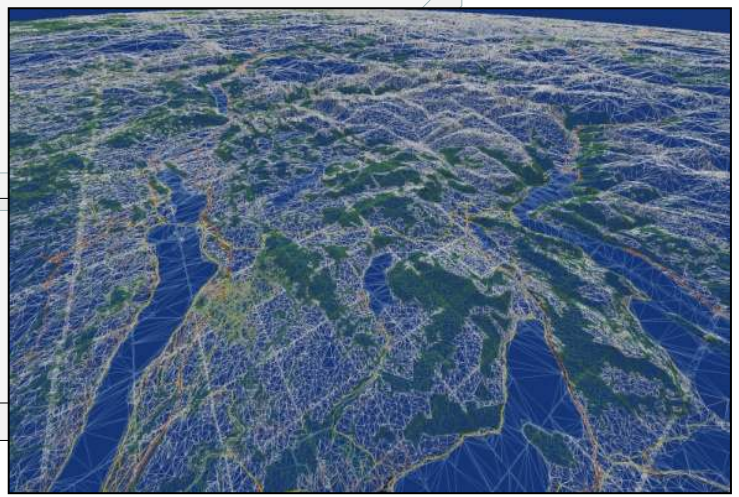
Spherical Mercator (Map Projection)

Cartesian Coordinates (Globe)



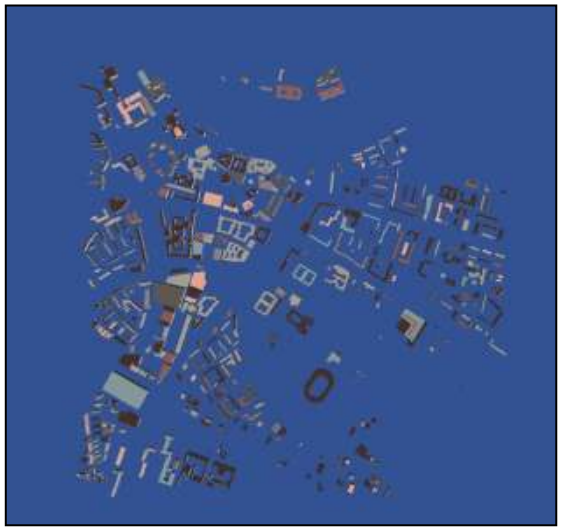
LOD Implementation





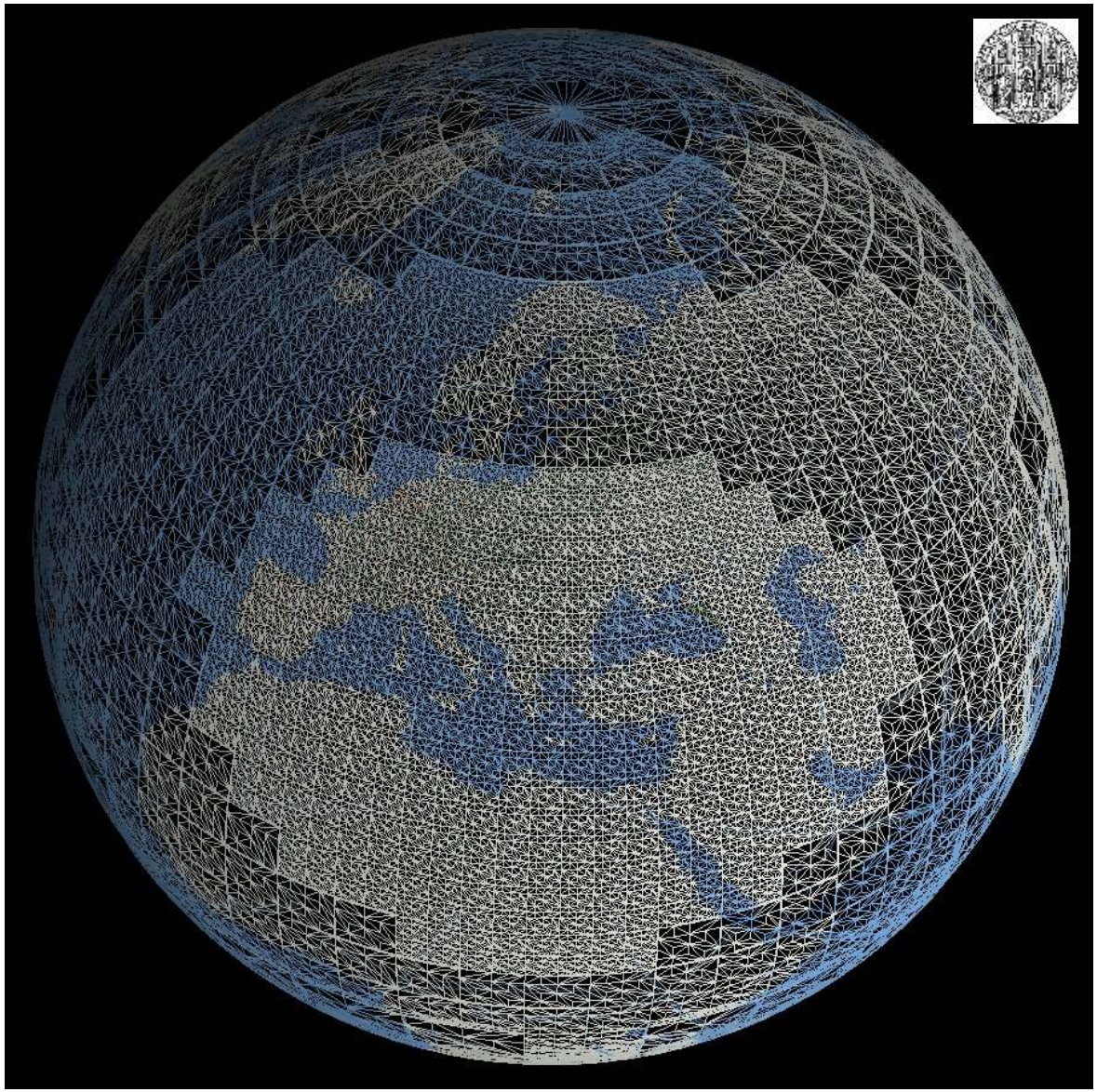
Terrain model has multiple levels

Objects are loaded at the base levels



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LOD Implementation



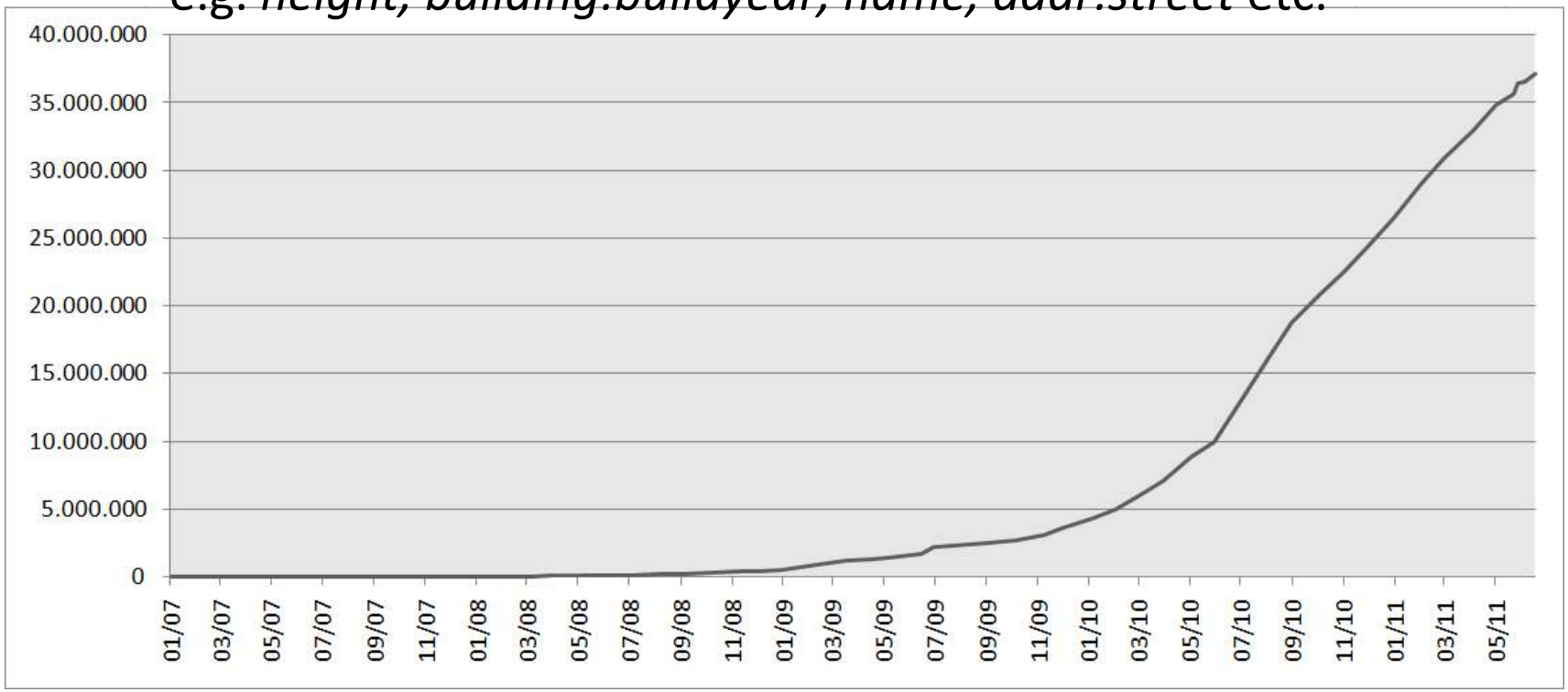
Tiles loaded into client

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Buildings in OSM

- more and more buildings (*building=yes*)
- Bing Maps
- Additional information

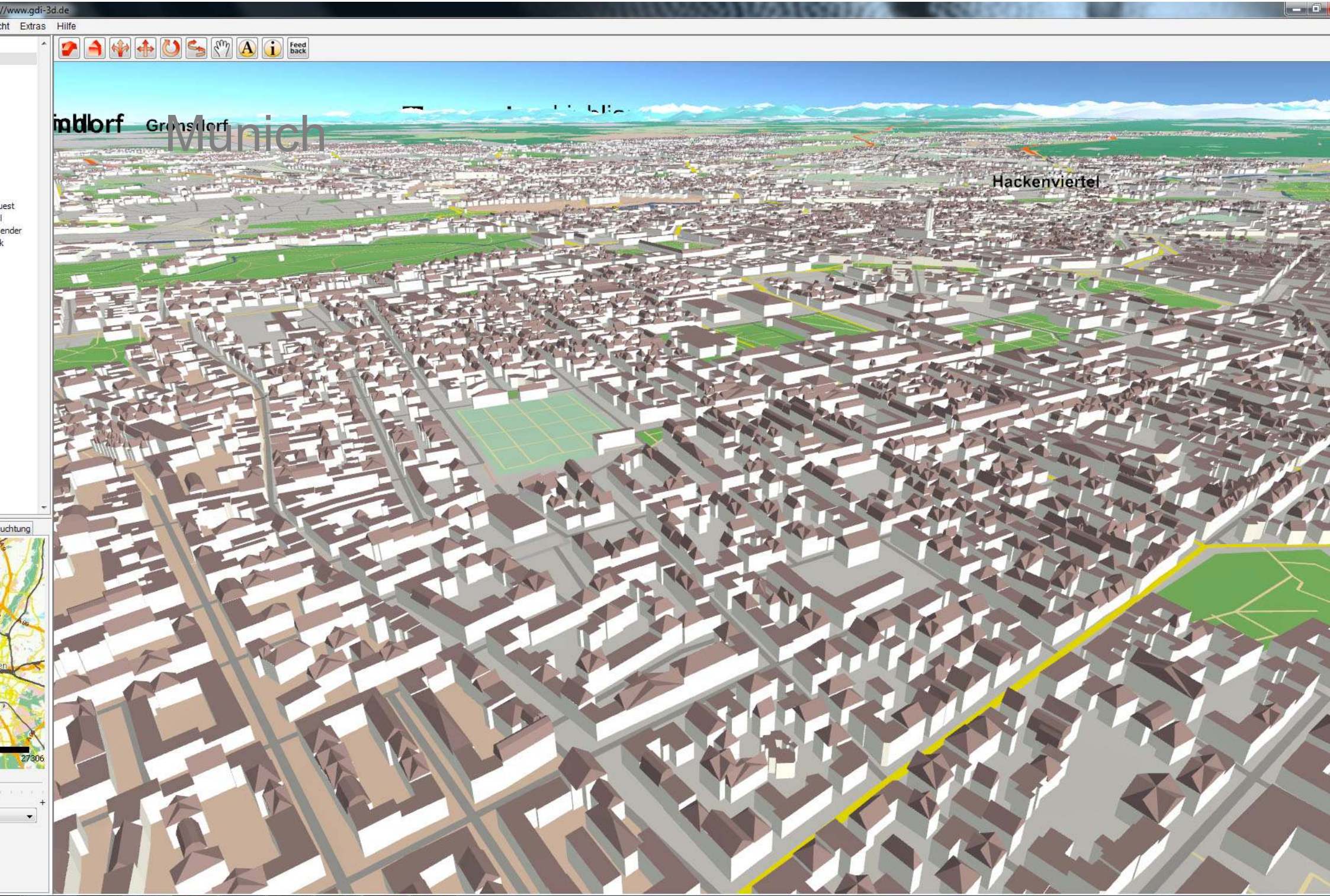
  e.g. *height, building:buildyear, name, addr:street* etc.

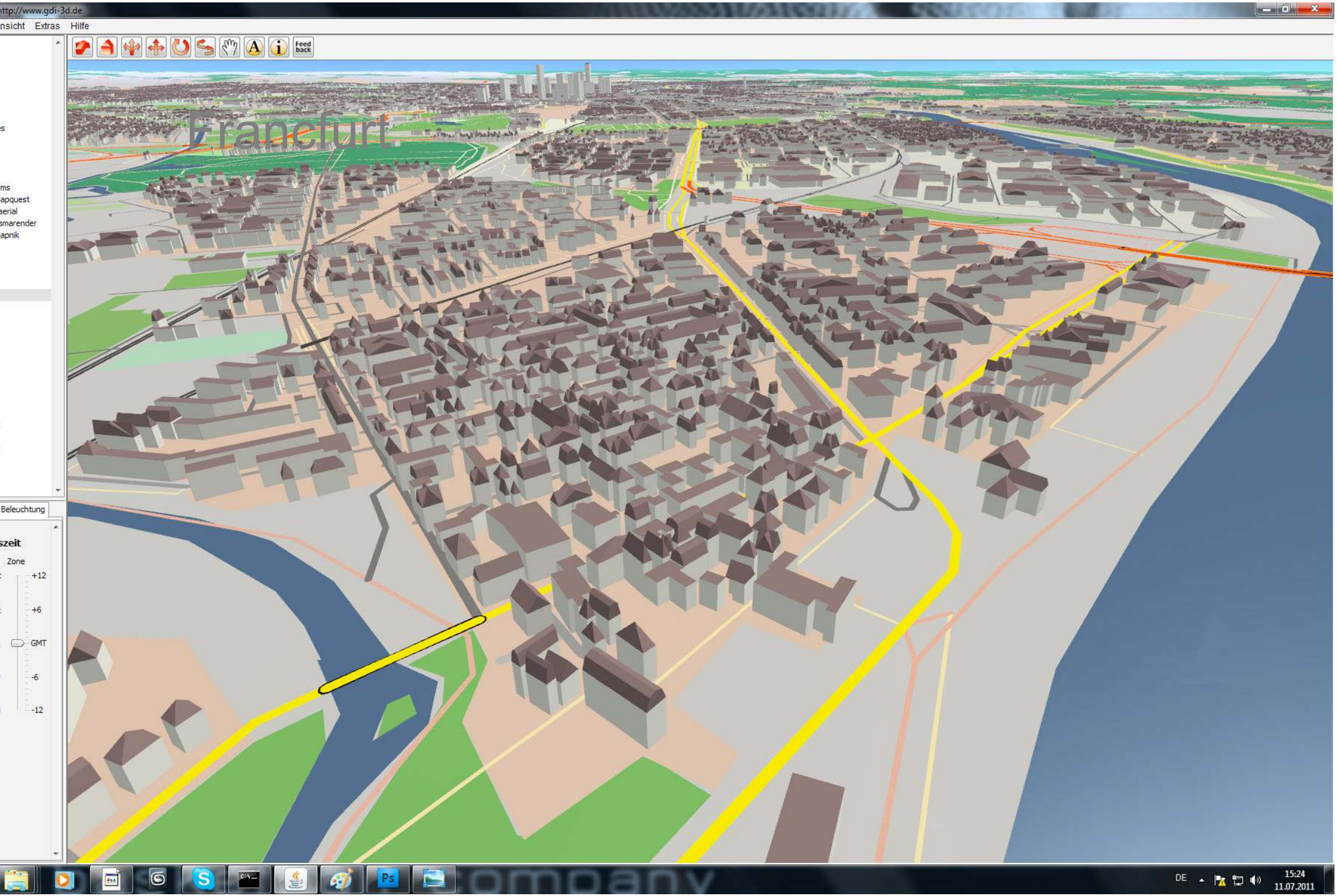


Modeling building:roofs

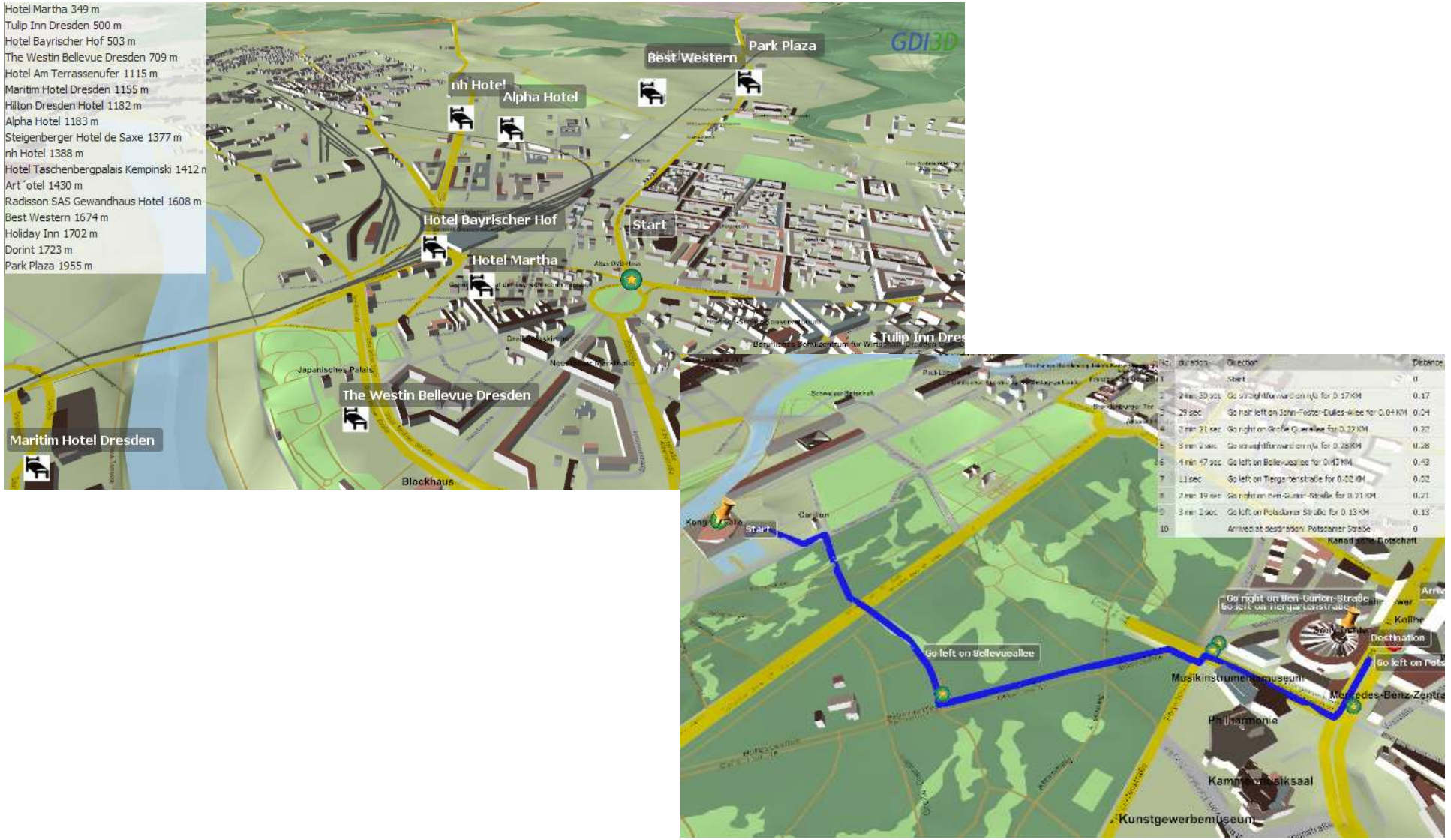
OSM key	Exemplary Value	Alternative
building:roof	tile, slate, flat, tile_red, reet	
building:roof:angle	30, 45, 10, 15	
building:roof:colour	grey, red, brown	building:roof:color
building:roof:material	shingles, slate, cardboard	
building:roof:orientation	along, across	
building:roof:extent	0.1, 0.3	
building:roof:ridge	yes	
building:roof:shape	pitched, hipped, flat, ridged	building:roof:style, building:roof:type

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Munich
Hackenviertel

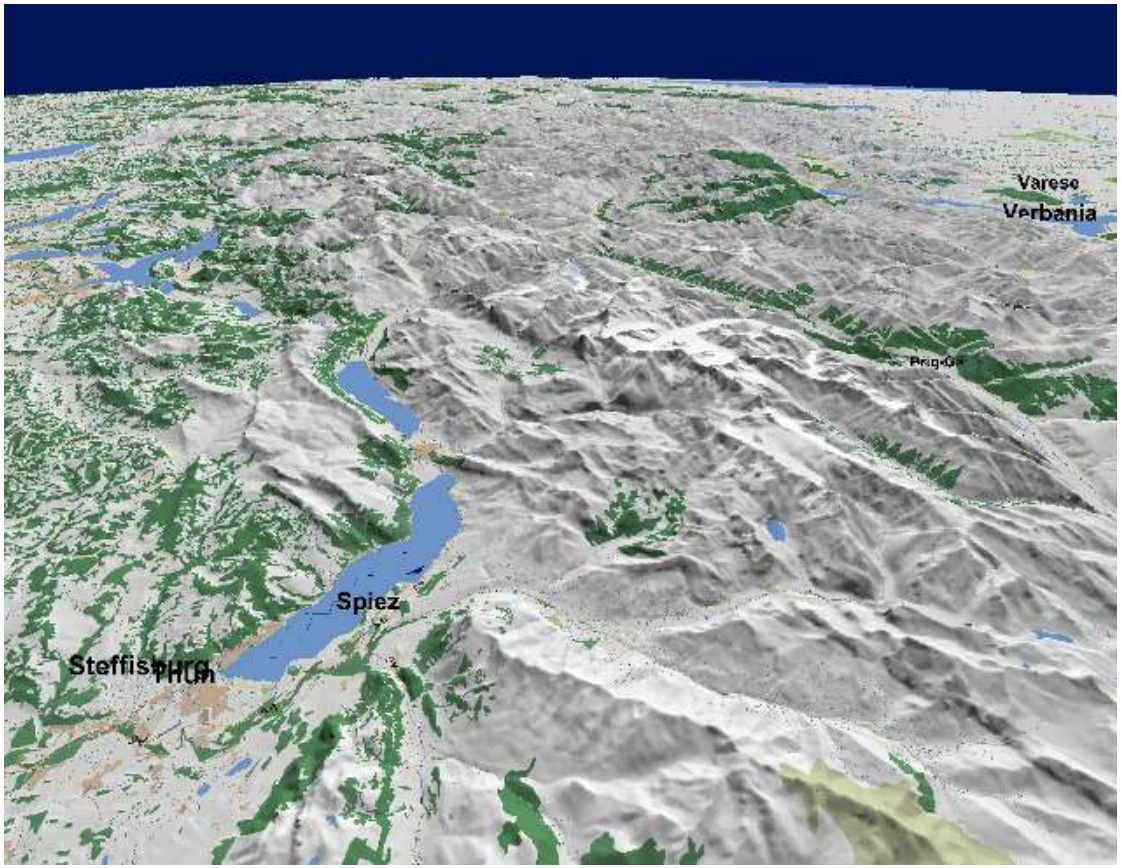

http://www.gdi-3d.de
nsicht Extras Hilfe
Feed back
Frankfurt
Beleuchtung
szeit
Zone
+12
+6
GMT
-6
-12
DE
15:24
11.07.2011

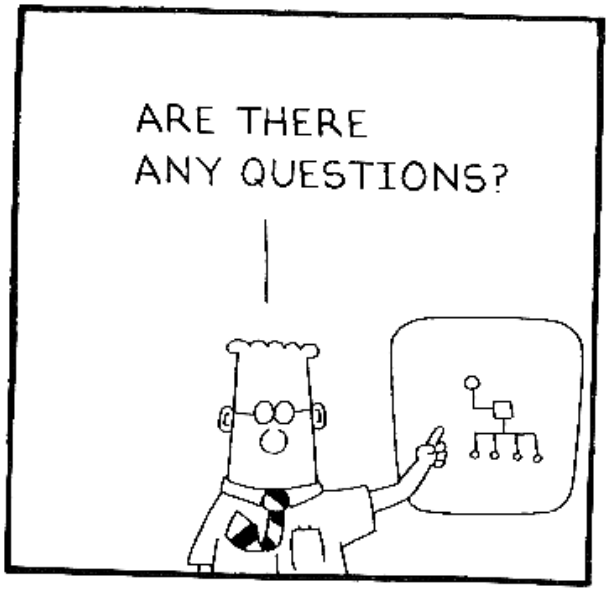
OpenLS local search/ Routing in OSM-3D



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Thank you very much!

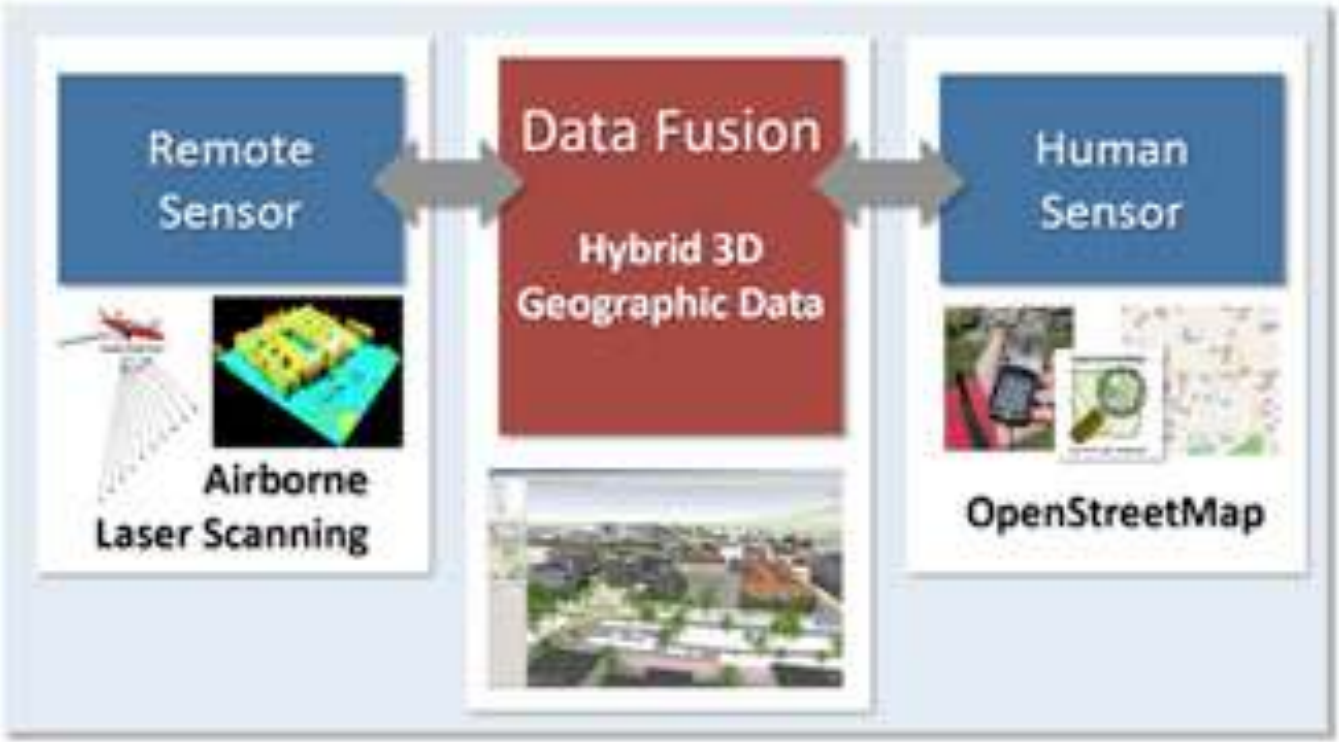




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Hengstberger Symposium 2011

- **Towards Digital Earth - 3D Spatial Data Infrastructures**
- Heidelberg, September 7-8, 2011
- http://www.geog.uni-heidelberg.de/lehrstuehle/gis/3dde.html



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